ALGEBRA 2 · UNIT 9: CONIC SECTIONS

Signals & Dishes

Use circles, parabolas, ellipses, and hyperbolas to design communication technology.

CP2

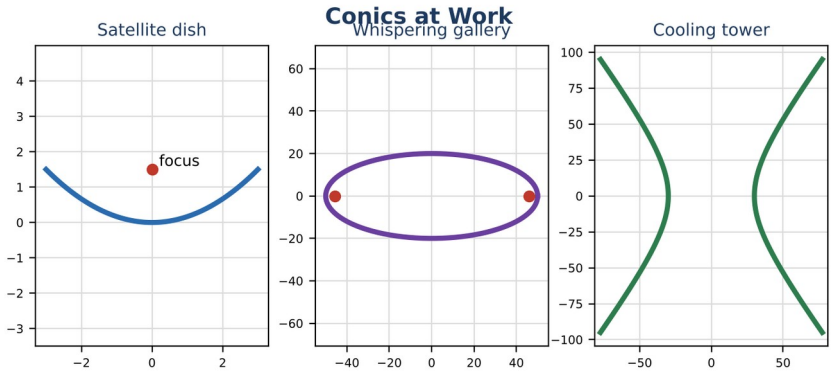
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

SignalWorks builds cell towers, satellite dishes, and even whispering galleries for science museums. Each design depends on one of the conic sections.

You will analyze each design with its equation, then create a coverage plan for a new town.



Learning Goals

- I can identify features of conics from their equations.
- I can test whether a point is inside a circle.
- I can use the focus of a parabola.

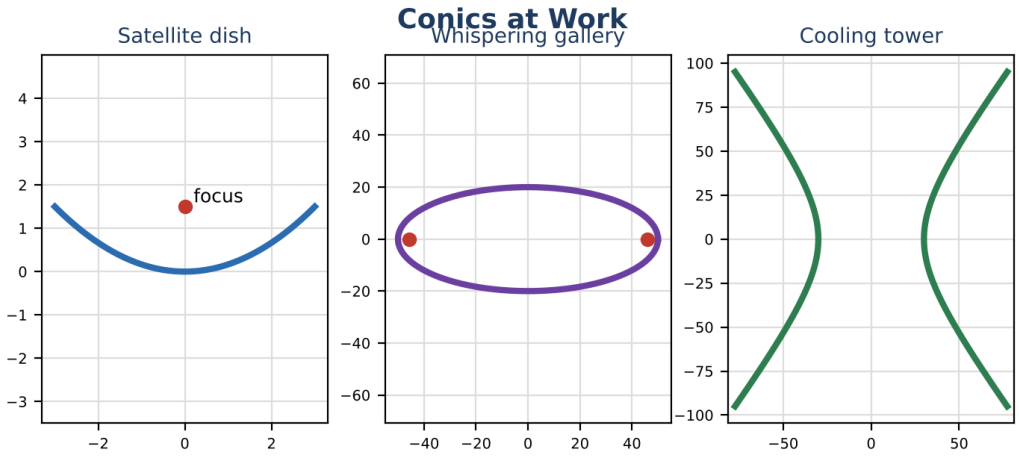
Project Timeline

When	What you will do
Day 1	Towers & Dishes
Day 2	Ellipses & Hyperbolas
Day 3	Coverage Map
Final	Engineering Design Brief + reflection

Part 1: Towers & Dishes

Day 1 · A tower covers the region inside $(x - 3)^2 + (y + 1)^2 = 16$ (miles).

1. Find the tower's location (center) and coverage radius.



2. Is a house at (6, 2) covered? Is one at (5, 1)? Substitute to check.

3. A satellite dish has the shape $x^2 = 6y$. For $x^2 = 4py$ the focus is at $(0, p)$. Where should the receiver go?

4. How deep is the dish if it is 6 ft across (x from −3 to 3)?

Part 2: Ellipses & Hyperbolas

Day 2 · Use the formulas.

5. A whispering gallery has the shape $x^2 \div 2500 + y^2 \div 400 = 1$ (feet). Find a, b, and $c = \sqrt{(a^2 - b^2)}$. Where should two people stand to hear whispers?

6. A cooling tower's side has the shape $x^2 \div 900 - y^2 \div 1600 = 1$. Find a, b, and $c = \sqrt{(a^2 + b^2)}$.

7. Match each conic to its real-world use and say why the shape is useful.

Part 3: Coverage Map

Day 3 · Plan coverage for a town with houses at (0, 0), (4, 2), (−3, 5), and (6, −4).

8. Place one or two towers (radius 5 miles each). Write each equation.

9. Test every house.

10. Draw the map.

Final Product: Engineering Design Brief

Your final product must include:

- Coverage map with circle equations
- Dish focus and depth
- Gallery and tower features
- Design brief

Reflection

R1. Why is the receiver placed at the focus?

R2. How does a circle equation come from the distance formula?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Coverage plan uses correct circle equations and point tests, and every conic feature is correct. Goes beyond the minimum with a creative, realistic choice.	Coverage plan uses correct circle equations and point tests, and every conic feature is correct.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Brief includes each conic's equation, key features (center, foci, vertices, radius), and a coverage map. Polished, organized, and visually engaging.	Brief includes each conic's equation, key features (center, foci, vertices, radius), and a coverage map. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.